Supporting Information: Supporting Information: When a Learned σ-Profile Is Not One: Diagnosing the Intermediate of a Physics-Embedded Solubility Model

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This document supports the article in five parts, in the order its Supporting Information section lists them: the architecture and the training schedule (§1); the data, the splits and the auxiliary supervision streams (§2); the reading rule declared for each measurement before it was made, with its outcome (§3); what the observations leave undetermined, which is the premise the measurements rest on (§4); and a second system in which a linear free-energy relation stands in for the closure (§5). Every number here is produced by a named script in the deposited repository from a named file in it; the mapping is `MANIFEST.md` in that repository.

1 Architecture and training

The model. A shared message-passing encoder embeds the solute and the solvent graphs: hidden width 64 over 3 message-passing layers on 35-dimensional node and 8-dimensional edge features, with a two-layer residual role-specific branch. A set2set readout with 3 processing steps pools each graph, the two pass through a three-layer cross-attention block, and the 512-dimensional pair representation $[\,g_s,\, g_v,\, g_s \odot g_v,\, |g_s - g_v|\,]$, with dropout 0.012, feeds both heads. The model has 5.40M trainable parameters.

The crystal head predicts $T_m = 100 + 600\,\mathrm{logistic}(\cdot) \in [100, 700]\,\mathrm{K}$ and $\Delta H_{\mathrm{fus}} = S_H\,\mathrm{softplus}(\cdot)$, giving the ideal term $\Phi(T)$ of equation (1); the group-contribution T_m prior is affine-calibrated on the training split before entering the head. The σ-profile head is a feed-forward network on the same pair representation, ending in a normalised non-negative vector of 51 bins.

The repository carries two further encoder backbones, a graph transformer and a variant that splits messages into dispersive, polar and hydrogen-bonding channels. Neither is used by any run reported here, and no accuracy comparison between backbones is claimed.

The closure, and what it omits. The differentiable COSMO-SAC layer uses the standard VT-2005 / Lin–Sandler constants: effective segment area $a_{\mathrm{eff}} = 7.5\,\text{Å}^2$, electrostatic-misfit constant $\alpha' = 16466.72\,\mathrm{kcal\,Å^4\,mol^{-1}\,e^{-2}}$, hydrogen-bond coefficient $c_{\mathrm{hb}} = 85580$ with cutoff $\sigma_{\mathrm{hb}} = 0.0084\,\mathrm{e\,Å^{-2}}$, coordination number $z = 10$, and Staverman–Guggenheim normalisers $r_0 = 66.69\,\text{Å}^3$, $q_0 = 79.53\,\text{Å}^2$. The σ-profile

grid is 51 bins on $[-0.025, 0.025]\,\mathrm{e\,\AA^{-2}}$.

The layer runs the residual, segment-activity part of COSMO-SAC alone. The Staverman–Guggenheim combinatorial term, the contribution molecular size and shape make to the activity coefficient, is omitted, because it is a function of cavity volume and area rather than of the profile and so carries no information about the quantity under study. A substitution that replaces the profile and leaves the combinatorial term in place would move both terms and the attribution would not be clean.

Two fixed points, and the iteration counts they are run at. The pipeline contains two nested fixed points, each with its own iteration count, and the counts differ between training and evaluation.

The segment-activity fixed point is damped at $\lambda = 0.7$ and run for 16 iterations at training time and 30 at evaluation. The solid-liquid equilibrium fixed point is the damped map $x_2^{(k+1)} = \lambda \exp(-\Phi(T) - \ln \gamma_2(x_2^{(k)}, T)) + (1-\lambda)x_2^{(k)}$, run for 5 iterations in training and 20 at evaluation, at the same λ, which is one configuration field read by both loops.

A count is a free parameter between fitting and scoring, so a checkpoint scored at a count far from the one it was fitted at is evaluated through an operator it was never fitted against. For the checkpoints reported here the exposure is bounded directly. Sweeping the segment count over 8, 16, 30, 100 and 300 with weights and rows held fixed moves the seed-mean $\ln x_2$ MAE to 1.9155, 1.9195, 1.9275, 1.9362 and 1.9364, an interval 0.021 wide in which converging the solver makes the arm slightly *worse*. The substituted arm sweeps through the same code on the same weights and rows, giving 2.2300, 2.3061, 2.3356, 2.3370 and 2.3368 and a contrast of +0.3145, +0.3866, +0.4081, +0.4008 and +0.4004. Between the evaluation count and the fixed point the contrast moves −0.008, 1.9% of itself, and that movement is not a direction: three of five seeds carry its sign, the per-seed range $[-0.031, +0.002]$ spans zero, and one seed carries almost all of it.

The aggregate is insensitive because the per-row shifts cancel, not because they are small. At 30 iterations the median row moves 5.9×10^{-4} against the converged solve, its ninety-fifth percentile 0.136 and its worst row 2.03 in $\ln x_2$. The two profiles also converge at different rates, in the direction the asymmetry predicts: at 30 iterations the segment residual is a median 9.7×10^{-4} with a ninety-fifth percentile of 0.038 and a worst case of 0.073 in $\ln \gamma_2^{\infty}$ on the tabulation, against 2.4×10^{-3}, 0.077 and 0.162 on the learned profile, so the broader learned profile is about twice the slower wherever the residual is large. What the count does reach is undercounting: at eight iterations the contrast reads +0.31, understating it by 0.094. The exposure is therefore a head scored *below* the count it was fitted at, which is not the configuration of any run reported here.

Optimisation and curriculum. Training is a fixed three-phase curriculum: 30 epochs of auxiliary-property pretraining, 70 of full solid-liquid equilibrium training, and 10 of low-rate stabilisation, at learning rates 2.6×10^{-4}, 8.5×10^{-5} and 8.5×10^{-6}. The optimiser is Adam with weight decay 5×10^{-4}, gradient clipping at 2.0 and batch size 64.

The direct control shares the encoder, interaction and readout verbatim and emits $\ln x_2$ directly, with a thermometer encoding of temperature in place of the dependence the physics arm receives through $\Phi(T)$ and the closure. It was tuned separately: dropout 0.174 against 0.012, second-phase learning rate 8.6×10^{-4} against 8.5×10^{-5}, weight decay 4.2×10^{-5} against 5×10^{-4}, gradient clipping 2.54 against 2.0, over 110 epochs equal to the physics arm's total budget at the same width and depth. Width, depth, batch size and epoch budget are matched and the optimisation is not, so any difference between the two arms is a difference between two separately tuned pipelines and is not attributable to the thermodynamic bottleneck alone. A hyperparameter-matched control was not run.

Objective. The objective is a weighted sum. Its registry holds 32 terms, of which 17 are hard-wired to zero and two more default to zero

through the configuration: disabled experimental scaffolding, released with the code so that a reader of the configuration can see what is off. Table S1 lists the terms carrying a nonzero default weight, together with the σ-profile earth-mover term, which is applied outside the registry when the grounding stream is on. Per-phase weight overrides are pinned in each run's resolved configuration, which is deposited with the checkpoint.

The activity-model variants that are not reported. The code implements two further activity models as controls: a fitted NRTL excess-Gibbs-energy model, and its symmetric one-parameter collapse in which the two interaction parameters are tied to a single predicted $\ln \gamma_2^{\infty}$. Neither is used by any measurement in the article, which concerns the σ-profile and therefore requires the profile-based closure. Their weights were not retained, and no comparison between them and the COSMO-SAC arms is made here: the arms were separately tuned, and a difference between separately tuned pipelines is not a property of the activity model.

2 Data, splits and the auxiliary supervision streams

Sources. Solubility is BigSolDB 2.0.[S1] Crystal T_m and ΔH_{fus} come from an open melting-point pool together with a group-contribution prior; melting-point tables are auto-detected as degrees Celsius or kelvin, because a double conversion of an already-absolute table shifts every T_m by $+273\,\mathrm{K}$ and T_m enters the ideal term directly. Reference σ-profiles are the VT-2005 compilation[S2] as redistributed with Ref.,[S3] ingested to 1432 compounds together with the COSMO cavity volume. Infinite-dilution activity coefficients are from ThermoML.[S4] The acid-dissociation data of §5 are the OPERA compilation,[S5] 7904 records, from which a uniform net-formal-charge filter leaves 1004 rows amenable to a Hammett treatment, 158 meta or para and 846 ortho or heteroaromatic.

The reference profiles are computed for a single conformer and protonation state. Mismatch between that state and the one a measurement was made in is error in the reference, and it is not signed: it displaces the quantity every departure in the article is measured against. The article therefore compares the learned profile with one tabulation throughout and makes no claim about the tabulation itself.

Coverage, and why the matched set is small. The set on which a learned profile and a reference profile can both be scored against a measured activity coefficient is limited by the intersection of the two, not by either alone. Of the ThermoML pairs at 298 K, only two solutes lack a VT-2005 profile while their solvent has one, so enlarging the profile database would not materially grow the matched set; the binding constraint is joint coverage. Matching by RDKit canonical SMILES on both components gives 60 pairs at 298 K over 41 solutes and 17 solvents. Matching by InChIKey instead admits 85 pairs, at the cost of importing a profile across a tautomer or protonation variant, which is the same mismatch named above; the article uses the stricter rule, so that the profile charged against a label is the one the pipeline would actually supply.

The split. The split is by Bemis–Murcko scaffold:[S6] molecules are grouped by ring system plus linker with side chains stripped, and no scaffold appears in more than one partition. Every test metric in the article is computed on the 5608 labelled test rows, which span 147 solutes.

The seeded split is not stable across versions of the data-preparation code. Regenerating it reshuffles which scaffolds land in the test partition, which orphans every checkpoint and every number measured on the previous split. All numbers in this work are therefore computed on a single pinned split, whose per-file SHA-256 each run manifest records and whose files are deposited with the checkpoints rather than regenerated.

Table S1: Loss terms with a non-zero default weight. The σ-profile earth-mover term is not a registry entry and is active only while the grounding stream is on. A further 17 registered terms are hard-wired to zero.

role	term	default wt.
task	solubility (bin-weighted Huber on $\ln x_2$)	1.0
crystal grounding	melting point T_m	0.3
	fusion enthalpy ΔH_{fus}	0.3
activity grounding	infinite-dilution $\ln \gamma_2^{\infty}$	0.5
	finite-composition $\ln \gamma_2$	0.5
	Hansen parameters	0.2
	σ-profile earth-mover	stream
regulariser	solubility monotonicity in T	0.1
	physics preference	0.1
	adaptive-correction residual	0.01
	NRTL τ	0.01
control coupling	direct-arm negative log-likelihood	0.2
	direct-arm regulariser	0.05
architecture	mixture-of-experts balance	0.02

The auxiliary supervision streams. An auxiliary stream is a sidecar table in the processed-dataset format whose rows are self-solvent, carry no solubility label, and switch on the mask of one target only. Two streams feed the runs reported here: a crystal stream carrying melting points, and a σ-profile stream carrying reference profiles. A stream builder applies two guards and reports both in its build summary. The first excludes any pool molecule whose Bemis–Murcko scaffold appears in the validation or test partition, and aborts rather than proceed if a split file is missing. The second reports how many pool molecules are not already labelled for that target in the training partition, by canonical SMILES, because a stream that is mostly duplicates tests nothing and the null it produces would be read as evidence that grounding does not help.

One build violated the first guard, and the exposure it leaves. A correct build of the σ-stream retains 1319 of the 1427 VT-2005 molecules that parse into the pool. The stream file was at one point rebuilt with the exclusion pointed at the splits of a small debug dataset, and that build carries 1425 rows, of which 106 have a scaffold in the real test and validation union. The overlap is not confined to scaffold relatives: 91 of those rows are themselves held-out solutes by canonical SMILES.

What a leak can act on is the labelled test set, and there the exposure is smaller than those counts suggest. Of the 147 labelled test solutes, 7 have their own reference profile in the defective stream and 8, those 7 among them, have a scaffold relative in it: 297 and 351 of the 5608 rows, 5.3% and 6.3%. For a scaffold split, solute coverage rather than row fraction is the load-bearing statistic. The correctly guarded build leaves none of either. The crystal stream's build on disk records the correct exclusion, so among the streams feeding these runs the defect is confined to the σ one.

Per-run stream provenance was not logged. The deposited artifacts therefore cannot certify which build fed which arm, and the disclosure above is the bound rather than an attribution. The runner now records the stream hash alongside the split hash so that future runs are auditable directly.

Compute and determinism. The five checkpoints of record were trained on one machine from one commit, at seeds 42 to 46; each manifest records the commit, the configuration, the seed and the environment. Retraining the same arm on a different GPU architecture does not reproduce a checkpoint to within seed spread, so a comparison of two checkpoints

should be read against the hardware each was trained on. Analysis is deterministic given a checkpoint and a seed, and every quantity reported in the article is a median or a range over the five seeds.

3 The control for each measurement

A contrast of this kind is interpretable only against something that would have come out differently had the instrument been responding to something else. This section states, for each row of Table 1, what that control is and what outcome from it would have refuted the reading. Each control is part of the measurement: the script that produces a deposit computes its control in the same pass, over the same rows and the same weights, so the two cannot drift apart. None of these is a pre-registration, with the one exception named at the end.

Does the observable determine the σ-profile? Control: the same Jacobian spectrum computed on random profiles carrying the same total area. The reading is that the deficiency belongs to the closure as a map rather than to the trained head, so it is the control's *agreement* that supports it. A control resolving materially more directions than the learned profile would refute it, because the restriction would then be something the head does. The learned profiles give a median of two directions of fifty for 99% of the information, range one to five; the random profiles give two.

Does the observable determine the crystal term? Control: the same Cramér–Rao bound recomputed with the melting temperature and the enthalpy of fusion supplied as independent labels. A bound already tight without them would refute the reading. It is $\pm 20\,\mathrm{kJ\,mol^{-1}}$ without and ± 1.7 with.

Does the head use the freedom? Control: the share of the molecule's cavity surface each charge-density window holds. A residual distributed in proportion to area, that is an enrichment near one in every window, would refute the reading that the departure is directed. Over the 71 molecules of the clean pool the enrichment is 3.97 in the donor window, 2.03 in the acceptor window and 0.88 in the nonpolar core as solute, and 3.78, 1.96 and 0.89 as solvent.

Is the output layer responsible? Two controls, because the claim is negative and a negative needs both. The first is the smallest bin mass the same head reaches anywhere, which bounds what the normalisation makes impossible; the second is an untrained head of the same architecture, which gives the pedestal the parameterisation produces before any data. A head unable to go far below its pedestal, or an untrained head whose pedestal matched the trained one, would refute the reading. The head reaches 10^{-11}, and the untrained pedestal is a median 0.571 against 0.02 to 0.03 trained.

Is the reference state preserved? Control: the same calculation with one profile supplied on both sides, which returns zero identically. A control returning anything other than zero would place the deviation in the closure rather than in the model, and would refute the reading. It returns zero exactly; the model's own two role profiles give a median $|\ln \gamma_2|$ between 0.012 and 0.044 across seeds, largest 0.80.

What does misplaced mass cost, and where? Control: the same per-bin sweep on random profiles of the same total area, which gives the edge-to-core ratio a smooth object produces with no chemistry in it. A real-profile ratio inside the random profiles' would refute the reading that the cost map is a property of the closure's chemistry. The real ratios are $21\times$ and $15\times$ on the two row sets against $1.6\times$ random.

Does the intermediate decompose? Control: a randomly chosen part of the grid of the same size as the part being substituted,

twenty draws per seed. If substituting a random part were never worse than substituting none, the parts would add and a per-bin attribution would be defined. A random third is worse than the fully learned profile in 20% of the 20×5 draws.

Where does the error that reaches the closure live? Controls: exact zeros written into the bins where the reference is zero, and a random third of the grid, twenty draws per seed. A zero-bin repair recovering a share comparable to the nonzero-bin repair would refute the reading that the consequential error is in the occupied bins. The nonzero repair recovers $+89\%$ of the gap and the zero repair -9%.

What does supplying the reference profile do? Control: the rows in which no profile is replaced because neither component is tabulated, which must not move. Movement there would mean the substitution was reaching rows it does not touch and would refute every number read from this contrast. Those rows move by less than 4×10^{-6}. The activity coefficients improve from 1.75 ± 0.06 to 0.76 and the solubility worsens by 0.43 ± 0.04 over 5274 rows.

The one measurement with a rule written beforehand. The reference-state test was prompted by a reviewer's observation that a role-dependent profile would break $\gamma(i \text{ in } i) = 1$. A rule was committed before the quantity was computed, six minutes before the result in the repository history, and it reads as follows. The existing closure validation cannot see the effect, because it supplies one tabulated profile to both sides and so tests the closure rather than the model; under equal profiles $\gamma = 1$ holds by construction. The model is built with a role-specific branch, so one molecule receives different profiles in the two slots, and the quantity to compute is therefore $\ln \gamma_2^{\infty}$ with the solute-role profile in one argument and the solvent-role profile in the other, at 298.15 K, with the layer's default configuration. The declared control was the one-profile calculation, which must return zero exactly; the declared refutation was that control returning anything else, in which case the deviation would be the closure's. Both parts were then computed, and both came out as declared.

4 What the observations leave undetermined

Every measurement in the article is a statement about what the network puts in a slot the data do not fix. This section gives the measurements behind that premise. Both are read against a control, because a rank statement on a smooth object is a claim about the object only if the same statement fails on a comparable object without the structure asserted.

The profile direction the closure can see. Fix the weights and the rows and perturb the σ-profile along each of the 51 grid coordinates in turn, recording the response of $\ln \gamma_2$. Over 25 solvents and five seeds the resulting Jacobian spectrum has two of fifty directions carrying 99% of the information, with a per-seed range of one to five. The control is the same spectrum computed on random profiles of equal total area, and it gives two as well.

That agreement is the result, not a failure of the test. The deficiency is a property of the closure as a map from a 51-bin histogram to one number, not something the trained head does: an observation of $\ln \gamma_2$ constrains a two-dimensional shadow of the profile whatever profile is supplied. The remaining coordinates are not determined by the solubility data at all, and what the head places in them is therefore not decided by fitting.

The resolved direction has a chemical locus as well as a rank. The dominant eigenvector concentrates on the extreme-$|\sigma|$ bins on which the closure's hydrogen-bonding term acts, so the coordinates the observation leaves free are the nonpolar and weakly polar interior of the profile. A complementary measurement at infinite dilution gives the same picture: the aggregate activity Fisher matrix $J^{\top}J$ with $J = \partial \ln \gamma_2^{\infty} / \partial \sigma$ is rank one, and the per-molecule Fisher that governs an individual profile has

participation ratio about 1.4 of 51 even at full solvent coverage.

Three limits travel with these numbers. The participation ratio is a concentration statistic and not an exact count of resolved directions. It must be read against a permuted control rather than absolutely, because finite differences decorrelate the redundant flat-profile coordinates and inflate it. And it is measured at infinite dilution, so its transfer to the finite-composition regime is not established here. None of this says that recovering the tabulated profile is impossible in principle; it says the solubility observations do not require it.

The crystal term. The same question for the other head has a closed-form answer. Fit the four parameters of the solid-liquid relation to a solubility series and the information matrix has condition number 2.3×10^5: the enthalpy of fusion is collinear with the temperature slope of the activity term, so a change in one is absorbed by a change in the other. The Cramér–Rao bound that follows puts no estimator of ΔH_{fus} from a solubility series alone inside $\pm 20\,\mathrm{kJ\,mol^{-1}}$, which is wider than the range the quantity takes across these compounds. Supplying the melting temperature and the enthalpy of fusion as independent labels reduces that bound to $\pm 1.7\,\mathrm{kJ\,mol^{-1}}$.

This is why the crystal properties are supervised externally and why the article's question is about the profile rather than the crystal term: the crystal term has an external label, and the profile coordinates the closure cannot see do not.

5 A second system: a linear free-energy relation

The article's measurements are made on one pipeline with one closure, so they cannot say whether the behaviour depends on the closure's adequacy. This section reports a second system built so that adequacy varies by chemical stratum and is known from structure alone.

Construction. The closure is a fixed Hammett linear free-energy relation, $g = \mathrm{p}K_{a,0} - \rho\,\sigma^{\star}_{\mathrm{H}}$, over a reaction series s, which fixes the per-scaffold Hansch–Leo–Taft constants $(\mathrm{p}K_{a,0}, \rho)$, and a substituent constant $\sigma^{\star}_{\mathrm{H}}$. A network predicts $\sigma^{\star}_{\mathrm{H}}$ from structure and the prediction is pushed through the fixed relation, which is the same arrangement as a σ-profile pushed through COSMO-SAC. The tabulated Hansch–Leo values are the external reference for the predicted coordinate, exactly as the VT-2005 table is for the profile. The constant carries its subscript throughout so that it is not confused with the screening-charge density of the article.

Assignment is mechanical and reads structure only. Four reaction series are tried in the fixed order benzoic acid, phenol, anilinium, pyridinium, and a record enters under the first whose ionizing group matches exactly once, at a ring atom lying in exactly one six-membered aromatic ring. Ring positions are typed by shortest ring-bond distance from that atom: one bond ortho, two meta, three para. The constant $\sigma^{\star}_{\mathrm{H}}$ is the sum over substituted positions of the tabulated Hansch–Leo value for that substituent at that position, from a fixed list of nineteen; an unsubstituted parent has $\sigma^{\star}_{\mathrm{H}} = 0$.

Why adequacy varies, and how the strata are defined. The two classical failure modes of a Hammett relation fall on either side of the distinction the article draws. Resonance saturation is a curvature in the substituent constant that a linear relation structurally cannot represent: a defect of the map. Ortho and steric field effects are absent from the tabulated constants altogether: information the intermediate does not carry.

The strata follow from that. A record whose every substituted position is meta or para with a tabulated substituent lies where the relation is close to exact. A record carrying an ortho substituent, or one outside the tabulated list, lies where it is not; both enter $\sigma^{\star}_{\mathrm{H}}$ as zero. The rule is a function of structure and never reads a residual, so the strata cannot be chosen by outcome.

Data and one uniform filter. The source is the OPERA compilation,[S5] 7904 records. A small number are supplied as pre-ionized microstates, inconsistent with a neutral-scaffold Hammett assignment, and they degrade the well-specified stratum severely. They are removed by one rule applied uniformly to both strata, net formal charge different from zero, which never references a residual and so cannot be circular, and which keeps net-neutral substituents such as nitro. The filter removes 5 of 163 rows in the well-specified stratum and 352 of 1198 in the other, leaving 1004 Hammett-amenable rows, 158 and 846.

The comparison, and the result. The physics arm, encoder to $\hat{\sigma}_{\mathrm{H}}$ to the fixed relation, is trained on $K = 20$ stratified within-scaffold splits, and the per-molecule error is paired against the deterministic reference arm, stratum by stratum. Where the relation is well specified the reference wins: learned MAE 0.48 ± 0.09 against 0.315 ± 0.07, a gap of -0.165, the reference better in 19 of 20 splits. Where it is not, the reference loses: learned 0.83 ± 0.07 against 1.62 ± 0.08, a gap of $+0.79$, the reference worse in 20 of 20 splits.

The sign of the effect of supplying the true intermediate therefore follows the adequacy of the relation it is supplied to, within one system, under one protocol, with the strata fixed by structure.

Three limits. The comparison is within distribution. The companion run on a Bemis–Murcko scaffold split is deposited beside it, and there nothing flips: both arms degrade and the learned arm degrades further, so the reference helps on both strata. The reason the main comparison is not run on a scaffold split is that such a split collapses the well-specified stratum to two ring systems. Every solubility result in the article is measured on a scaffold split, so this is the one place the protocol differs.

The resampling unit is the reaction series and only four exist, so the cluster bootstrap has 35 distinct outcomes and a finest atom of about 0.004. The interval endpoints quoted above are percentiles of that law and are no finer; read as a sign test on four units the two statements are $p = 1/16$ and $p \approx 0.31$. What the result rests on is the consistency of the sign across splits and series and the size of the gap in the misspecified stratum, $+0.79$ against a reference error of 1.62, and not on a P value. The refit that would settle the series-level question at four units was not run.

Finally, a competence sweep over training fractions 1.0, 0.5, 0.25 and 0.12, six seeds each, shows that the reference side of the comparison is fixed by the stratum and not by the data budget: its error is the same at every budget while the learned side moves. In the well-specified stratum the learned arm rises from 0.46 to 0.98 and never reaches the reference's 0.31, so the reference helps throughout. In the other it stays below the reference's 1.62 across the whole range, 0.85 at full data and 1.31 at 12%, so the reference hurts throughout. Only scaffold extrapolation, where the predicted constant breaks down to 2.6 ± 2.0, pushes it back above and reverts the sign, and that reversion is a property of the mean over six splits rather than of each: three of the six still have the reference hurting, and the negative mean is carried by two splits.

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